

Bhavesh Kukreja

[Mail](#) | [LinkedIn](#) | [Portfolio](#) | [GitHub](#) | [LeetCode](#)

EDUCATION

University of Mumbai – Thadomal Shahani Engineering College

Mumbai, India

Bachelor of Engineering in Computer Engineering (CGPA: 9.31/10.0)

2023 – 2027

EXPERIENCE

Backend Engineering Intern (Python)

December 2025 – February 2026

Zobyt

Remote

- Built a newsletter scheduling API in **FastAPI** with recurring tasks, user rate limits, and error rollback
- Reduced a data-sync script's runtime from 45 minutes to 4 by fixing **N+1 API calls** with metadata pre-fetching
- Resolved a **PostgreSQL** connection leak by refactoring the application's database session handling
- Fixed a scheduler bug bypassing rate limits by correcting a data model **serialization** error
- Improved CI/CD stability by refactoring flaky tests with deterministic database queries in the **Pytest** suite

PROJECTS

Rexus | *FastAPI, React, Redis, PostgreSQL, Docker* | [GitHub](#) | [Video Demo](#)

- Developed an API Gateway in **FastAPI** with key authentication, response caching, and rate limiting
- Implemented a **Redis** sliding window rate limiter and response cache to minimize API latency
- Built an **async logging system** with a **Redis** buffer and background worker to prevent I/O blocking
- Designed an analytics dashboard in React using **WebSockets** for live metrics
- Containerized the application stack for deployment using **Docker Compose**

King's Gambit | *React, Django, WebRTC, Docker, PostgreSQL, Redis* | [GitHub](#) | [Video Demo](#)

- Developed a full-stack, room-based chess platform using **React**, **Django**, and **PostgreSQL**
- Used **Django Channels** and **Redis** to implement WebSocket-based real-time gameplay
- Integrated **WebRTC** video chat using **PeerJS** for P2P communication to reduce server bandwidth
- Designed a multi-service architecture and containerized the application with **Docker Compose**
- Implemented security features including authentication, **CSRF** protection, and API rate-limiting

AutoTriage | *Python, FAISS, Gemini API, Ollama, Pydantic* | [GitHub](#)

- Built a terminal-based **RAG** pipeline to route, classify and answer support tickets across 3 product domains
- Implemented **regex pre-routing** to bypass the LLM entirely for adversarial and out-of-scope tickets
- Enforced structured output with **Pydantic** and prevented prompt injection using **XML-sandboxed** inputs
- Supported Gemini 2.5 Flash and Qwen 2.5 via **Ollama** with defensive JSON parsing

EXTRACURRICULAR ACTIVITIES

Technical Writing

- Analyzed **CPython** source code to explain how lists use **dynamic C arrays** for performance | [Article](#)
- Explained **dictionary structure** in CPython detailing its **O(1) lookups** and insertion order | [Article](#)
- **Competitive Programming:** Solved **200+** Data Structure & Algorithms problems on [LeetCode](#)
- **Hackathons:** Ranked in the **top 10%** (132/1349) in **Hackerrank Orchestrate** May 2026
- Initiated and manage a tech community with **450+** members for discussions on programming and tech trends

TECHNICAL SKILLS

Languages: Python, C, JavaScript

Backend: FastAPI, Django, Flask, Django Channels

Frontend: React, HTML, CSS, Tailwind

Databases: PostgreSQL, MySQL, SQLite, Redis

Tools & Technologies: Docker, Alembic, WebRTC, PeerJS, Git, GitHub

CERTIFICATIONS

Harvard CS50x – Introduction to Computer Science

Harvard CS50P – Introduction to Programming with Python